Implicit Learning of Professional Skills through Immersive Virtual Reality

Bondesan, P., Canal, A., Fleury, S., Boisadan, A., Richir, S. (2025). *Implicit Learning of Professional Skills through Immersive Virtual Reality: a Media Comparison Study.* Proceedings of the 2025 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 442–449

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Why This Paper Matters for HCI

Examines how immersive VR (HMD-based IVR) shapes cognition, motivation, and performance compared to conventional screen-based learning

- Connects to presence, embodiment, and user experience in VR learning systems
- Provides empirical evidence: motivation gains ≠ performance gains

Our motivation: How effective are VR headsets for teaching professional skills?



Research Context and Gap

Field

Immersive learning technologies, virtual reality, media comparison studies

Problem

Most VR research targets explicit learning with weak controls; little known about implicit, experiential learning

Context

Professional skills rely on tacit knowledge—hard to verbalize

Relevance

Should we invest in immersive VR for training?

Research Design

Participants

40 engineering students (25 women, 15 men, aged 19–23)

18 IVR, 22 slideshow

IVR Condition

Meta Quest 2 HMD

Move around virtual vine

30 trials, no narration

Slideshow Condition

Laptop presentation

30 fixed images

Same decisions, feedback

Task: Implicit learning of vine vigour assessment

Measures and Procedure

01

Pre-test

Demographics and prior knowledge

02

Lesson

Individual session (IVR or slideshow)

03

Motivation Survey

Intrinsic Motivation Inventory

04

Transfer Test

Real vineyard (1 week later)

Dependent Variables

- Accuracy: correct classifications (0–30)
- Time: total task completion
- Motivation: interest/enjoyment (7-point scale)

Design Validity

Content equity between media ensures immersion and interaction modality are main differences



Why This Design Works

Implicit Learning Strategy

Repeated perceptual judgments + feedback reduce explicit cognitive load

Ecological Validity

Authentic vineyard test connects lab interaction with real-world performance

World-Centered Interaction

Aligns with experiential learning—tacit knowledge over verbal rules

Key Findings

17.2

IVR Accuracy

Correct classifications out of 30

16.5

Slideshow Accuracy

Correct classifications out of 30

6.2

IVR Motivation

7-point IMI scale

5.5

Slideshow Motivation

7-point IMI scale

1

Accuracy: NS

No significant difference

2

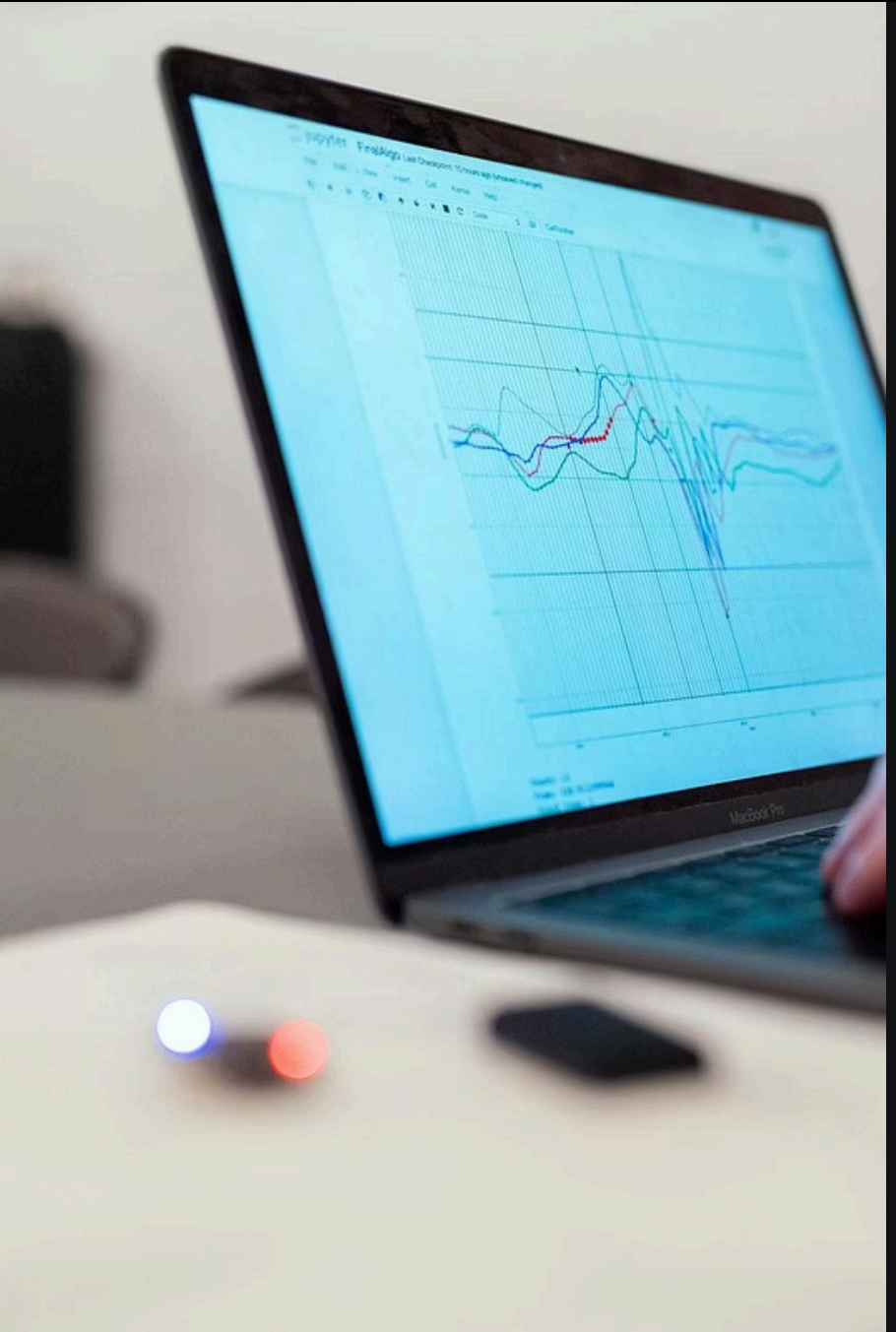
Time: IVR Slower

8.6 vs. 6.5 minutes

3

Motivation: IVR Higher

Significantly elevated



Limitations to Consider

Training Dosage

Only 30 trials may be insufficient for complex tacit skills

Metrics Gap

Time and accuracy miss process data: interaction logs, gaze patterns, qualitative insights

Ecological Complexity

Real vineyards vary widely—interface strategies may interact with environmental factors

Conclusions

No Accuracy Advantage

IVR did not improve decision accuracy vs. slideshow

Slower Decisions

IVR group took longer in real-world evaluation

Higher Motivation

Significantly elevated intrinsic motivation with IVR

> Immersive systems benefit engagement more clearly than immediate performance for cognitive professional skills



Future Research Directions

Spatial Tasks

Explore tasks leveraging spatial reasoning, motor skills, social interaction

Hybrid Learning

Combine implicit and explicit strategies for intuitive pattern acquisition + targeted guidance

Critical Analysis

Strengths

- Clear media-comparison design with content-equivalent conditions
- Real-world transfer task
- Focus on implicit learning—fills important research gap

Weaknesses

- VR interaction limited—no vine manipulation
- Missing presence, embodiment, workload measures

Key takeaway: More immersive interfaces increase motivation without guaranteeing better task performance

Future potential: VR shows promise for engagement-driven training contexts where motivation is paramount

References

- **Original paper:** Bondesan, P., Canal, A., Fleury, S., Boisadan, A., Richir, S. (2025). Implicit Learning of Professional Skills through Immersive Virtual Reality: a Media Comparison Study. *Proceedings of the 2025 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 442–449. **https://ieeexplore.ieee.org/document/10937417**
- **Video: https://www.youtube.com/watch?v=D3bXNpgRHPs**